

PART A: MULTIPLE CHOICE QUESTIONS

Instructions: Select the best answer for each question. (1 point each)

1. What percentage of a data scientist's time is typically spent on non-analytical tasks such as ETL, data cleaning, and preprocessing?

- ☐

A) 20%

- ☐

B) 40%

- ☐

C) 60%

- ☐

D) 80%

2. Which NVIDIA library provides GPU acceleration for pandas with zero code change?

- ☐

A) cuML

- ☐

B) cuDF

- ☐

C) Triton Inference Server

- ☐

D) XGBoost

3. What is the internal data format used by cuDF for efficient processing of tabular data?

- ☐

A) JSON

- ☐

B) CSV

- ☐

C) Apache Arrow columnar format

- ☐

D) Parquet

4. Which of the following is NOT one of the three pillars of machine learning according to the workshop?

- ☐

A) Data

- ☐

B) Algorithm

- ☐

C) Visualization

- ☐

D) Compute

5. What technique is used to protect against overfitting during model evaluation?

- ☐

A) Grid search

- ☐

B) Cross validation

- ☐

C) Pruning

- ☐

D) Ensemble learning

6. Which graph representation shows the smoothness of the graph?

- ☐

A) Adjacency matrix

- ☐

B) Degree matrix

- ☐

C) Laplacian matrix

- ☐

D) Adjacency list

7. What is the primary advantage of lazy execution in Polars?

- ☐

A) Faster single-row operations

- ☐

B) Global query optimization and reduced memory usage

- ☐

C) Automatic CPU fallback

- ☐

D) Real-time data processing

8. Which algorithm is described as the "gold standard in single model performance for classification and regression"?

- ☐

A) Decision Trees

- ☐

B) Random Forest

- ☐

C) XGBoost

- ☐

D) K-Means Clustering

9. What does the "split-apply-combine" paradigm refer to?

- ☐

A) Data visualization process

- ☐

B) GroupBy operations

- ☐

C) Model training workflow

- ☐

D) Graph representation

10. Which tool is designed for deploying AI models at scale with multi-framework support and dynamic batching?

- ☐

A) cuDF

- ☐

B) Triton Inference Server

- ☐

C) Dask

- ☐

D) nx-cugraph

11. What is the expected growth factor for data processing needs by 2025 according to the workshop?

- ☐

A) 100X

- ☐

B) 500X

- ☐

C) 1000X

- ☐

D) 10,000X

12. Which type of analysis answers the question "What will happen?"

- ☐

A) Descriptive Analysis

- ☐

B) Diagnostic Analysis

- ☐

C) Predictive Analysis

- ☐

D) Prescriptive Analysis

13. What is the key advantage of the Apache Arrow columnar format?

- ☐

A) Faster row-by-row processing

- ☐

B) Zero-copy data transfer

- ☐

C) Smaller file size

- ☐

D) Better encryption

14. In graph analytics, what does "1-hop" refer to?

- ☐

A) One edge from the source

- ☐

B) Direct neighbors

- ☐

C) The shortest path

- ☐

D) The degree of a node

15. Which of the following is a use case for clustering?

- ☐

A) Classification

- ☐

B) Market segmentation

- ☐

C) Regression

- ☐

D) Feature selection

PART B: TRUE OR FALSE

Instructions: Circle T for True or F for False. (1 point each)

16. Vectorized operations are typically slower than equivalent loops or iterations.

T / F

17. DLPack and cuda_array_interface are standard formats that enable sharing GPU arrays between different libraries without copying or converting data.

T / F

18. In unsupervised learning, models are trained on labeled data to make predictions on new, unseen data.

T / F

19. Feature engineering is considered one of the most crucial and impactful steps in the machine learning process.

T / F

20. The workshop covers statistical analysis, neural networks, and distributed computing in detail.

T / F

21. XGBoost uses regularization to prevent overfitting and improve model generalization.

T / F

22. Polars uses eager execution by default, making it faster for large datasets.

T / F

23. The Laplacian matrix is calculated as: Degree - Adjacency.

T / F

24. Pruning is a technique used to prevent overfitting in decision trees.

T / F

25. RAPIDS can provide up to 150x faster performance than CPU-only pandas.

T / F

PART C: SHORT ANSWER QUESTIONS

Instructions: Provide brief but comprehensive answers. (5 points each)

26. What are the four types of analysis in data science? Briefly explain each.

27. List and explain three ways to utilize nx-cugraph for GPU-accelerated graph analytics.

28. Describe the six key components of the Machine Learning Model Development Workflow.

29. Explain the key differences between supervised and unsupervised learning.

30. What is the RAPIDS ecosystem and what is its purpose in data science?

ANSWER KEY

NVIDIA FUNDAMENTALS OF ACCELERATED DATA SCIENCE

Final Assessment Quiz - Answer Key

PART A: MULTIPLE CHOICE ANSWERS

Question	Answer	Explanation
1	B) 40%	Data scientists spend 40% of their time on non-analytical tasks
2	B) cuDF	cuDF accelerates pandas with zero code change (150x faster)
3	C) Apache Arrow columnar format	cuDF uses Apache Arrow as its internal data format
4	C) Visualization	The three pillars are Data, Algorithm, and Compute
5	B) Cross validation	Cross validation is used to protect against overfitting
6	C) Laplacian matrix	Laplacian matrix shows graph smoothness (Degree - Adjacency)
7	B) Global query optimization	Lazy execution enables optimization and reduced memory usage
8	C) XGBoost	XGBoost is the "gold standard" for classification and regression

Question	Answer	Explanation
9	B) GroupBy operations	GroupBy follows split-apply-combine paradigm
10	B) Triton Inference Server	Triton is designed for scalable AI model deployment
11	C) 1000X	Data processing needs are expected to grow 1000X by 2025
12	C) Predictive Analysis	Predictive analysis answers "What will happen?"
13	B) Zero-copy data transfer	Arrow enables zero-copy data transfer between tools
14	B) Direct neighbors	1-hop refers to direct neighbors of a node
15	B) Market segmentation	Clustering is used for market segmentation

PART B: TRUE/FALSE ANSWERS

Question	Answer	Explanation
16	FALSE	Vectorized operations are MUCH faster than loops
17	TRUE	DLPack and cuda_array_interface enable GPU array sharing
18	FALSE	Supervised learning uses labeled data; unsupervised does not
19	TRUE	Feature engineering is crucial for model performance
20	FALSE	The workshop explicitly excludes these topics

Question	Answer	Explanation
21	TRUE	XGBoost uses regularization to prevent overfitting
22	FALSE	Polars uses LAZY execution by default
23	TRUE	Laplacian = Degree - Adjacency
24	TRUE	Pruning prevents overfitting in decision trees
25	TRUE	RAPIDS cuDF can be 150x faster than CPU pandas

PART C: SHORT ANSWER ANSWERS

Question 26: Four Types of Analysis

Type	Question	Explanation
Descriptive Analysis	What happened?	Summarizes historical data to understand past events
Diagnostic Analysis	Why it happened?	Examines data to understand causes of events
Predictive Analysis	What will happen?	Uses data to forecast future outcomes
Prescriptive Analysis	What can be done?	Recommends actions based on analysis results

Question 27: Three Ways to Utilize nx-cugraph

1. Environment Variable Method

- Set NX_CUGRAPH_AUTOCONFIG environment variable
- NetworkX automatically dispatches to backends

- Also works in Jupyter Notebooks with %env

2. Backend Keyword Argument Method

- Explicitly specify backend='cugraph' in NetworkX APIs
- Provides direct control over backend selection
- Clear and explicit usage

3. Type-based Dispatching Method

- Import the desired backend
- Create a Graph instance for the backend
- Ensures specific behavior without runtime conversions

Question 28: Six Components of ML Model Development Workflow

Component	Description
1. Data Preparation	Clean, transform, and prepare data for analysis
2. Feature Engineering	Transform raw data into better representations; often the most crucial step
3. Model Selection	Choose appropriate algorithms based on the problem and data
4. Model Training	Fit the model to the dataset
5. Model Evaluation	Assess model's performance using various metrics
6. Model Deployment	Integrate the model into production systems

Additional Considerations:

- Define business problem first
- Gather data from relevant sources
- Perform exploratory data analysis (EDA)

- Continuously monitor and retrain as needed

Question 29: Supervised vs Unsupervised Learning

Aspect	Supervised Learning	Unsupervised Learning
Definition	Trained on labeled data to make predictions or classifications	Identifies patterns without predefined categories
Input	Features + Labels/Targets	Features only (no labels)
Goal	Learn mapping from input to output	Discover hidden patterns and groups
Examples	Classification, Regression	Clustering, Dimensionality Reduction
Data Requirements	Requires labeled data	Can work with unlabeled data
Use Cases	Spam detection, Medical diagnosis	Market segmentation, Anomaly detection

Question 30: RAPIDS Ecosystem and Purpose

Definition: RAPIDS is an optimized hardware-to-software stack for the entire data science pipeline.

Purpose:

1. **GPU Acceleration** - Provides GPU-acceleration for ETL, analytics, and machine learning
2. **Seamless Integration** - Enables seamless integration across data science workflows
3. **Productivity** - Reduces processing time and improves productivity

4. **Interoperability** - Uses shared, standard data structures for interoperability
5. **Efficiency** - Minimizes data movement between CPU and GPU
6. **Innovation** - Enables more experimentation and model tuning

Key Components:

- **cuDF** - GPU DataFrame library for data manipulation
- **cuML** - GPU-accelerated machine learning algorithms
- **cuGraph** - GPU-accelerated graph analytics
- **cuXfilter** - GPU-backed data visualization

Overall Impact:

- Accelerates the entire data science pipeline
- Reduces time to insight and total cost of ownership
- Fosters data-driven decision making
- Provides up to 150x speed improvements